Bhadra Korukonda

AI Engineer
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Detail-oriented Computer Science student with practical experience in AI and Machine Learning. Developed graph-based knowledge retrieval systems (GRAIL-LM) integrating GNNs and LLMs, and NLP-driven resume ranking models. Skilled in Python, data analysis, scalable deployment, and model performance optimization techniques.

SKILLS

Programming Languages: Python, Java, Javascript
Libraries/Frameworks: Pytorch, Scikit,Numpy,Flask, React
Tools/Platforms: FastAPI, Flask, REST APIs, Git, Docker,Kubernetes, Promotheus, Graphana
Databases: PostgreSQL, Mongo DB, SQL

PROFESSIONAL EXPERIENCE

Research Centre Imarat(RCI), DRDO
Datascientist Intern | Python, SQL, Git, PyTest, Automation — April 2025 – Present

- Automated Git workflows with Python, cutting version control overhead by **30%**.
- Developed statechart validation tools using OOP and tests, boosting accuracy by **25%**.
- Streamlined pipelines with SQL and Git CI, accelerating debugging and deployment.

IIT Hyderabad
Research Intern | Python, PyTorch Geometric, Pandas, Geopy, GRU/LSTM, GNNS(GCN, GAT, EGAT) — June 2024 – Aug 2024

- Implemented a hybrid Edge-Aware Graph Attention Network (EGAT-GRU) to forecast PM2.5 levels across Delhi using spatio-temporal air quality data.
- Encoded wind direction and geodesic distance as edge features, enabling wind-informed pollutant transport learning and dynamic spatial attention.
- Achieved state-of-the-art performance (RMSE = 16.72, MAE = 10.05), outperforming GAT, GCN, and physics-based baselines for real-time pollution prediction.

PROJECTS

GRAIL-LM | FastAPI, LangChain, Neo4j,Docker, Ollama (LLaMA3)

- Developed an end-to-end Graph-RAG system integrating Large Language Models with Knowledge Graphs using FastAPI, LangChain, and Neo4j for explainable multi-hop question answering.
- Achieved ~25–30% improvement in factual accuracy and sub-2 s response latency by grounding LLM outputs in graph-structured data and deploying a containerized API-Streamlit pipeline.
- Engineered a production-ready AI stack (FastAPI + PyTorch Geometric + Docker) demonstrating scalable graph reasoning, LLM integration, and real-time visualization of reasoning paths.

Relfuse MVP | Pytorch, DGL,Transformers, GraphSAGE,FastAPI

- Developed a hybrid relational-feature fusion model combining GNN embeddings with transformer attention for structured data reasoning.
- Conducted feature ablation and hyperparameter tuning, achieving +18 % classification accuracy on benchmark graph datasets.
- Built modular inference APIs to serve predictions and visualizations, bridging research models with deployable analytics tools.

RESEARCH WORK

Alzheimer's Detection using Deep Learning | CNN, ResNet, CBAM, ADNI Dataset — Jan 2024 – May 2024

- Built an AI-powered plant identification system using **MobileNetV2** with **92% accuracy**.
- Implemented **OpenCV** preprocessing pipeline for real-time mobile image recognition and classification
- Integrated **Flask** API with **Gemini LLM** for generating plant-based medicinal insights and recommendations

Medical Text Simplification System | SimpleT5, MeDAL — Aug 2024 – Dec 2024

- Fine-tuned **T5-base transformer** using **SimpleT5** on the **MeDAL dataset** to simplify medical abbreviations and acronyms.
- Developed a translation pipeline with custom preprocessing algorithms and position-aware word replacement.
- Achieved **BLEU score of 0.87** and **ROUGE-L score of 0.84** on medical text translation tasks.

EDUCATION

Bachelor of Technology (CS Engg), Woxsen University, GPA: 3/4 — Aug 2022 – July 2026
Narayana Jr College(SSC), Hyderabad, Percentage: 87.1% — June 2020– May 2022
Chinmaya Vidyalaya(CBSE),Hyderabad Percentage: 90% — May 2008 – June 2020